



Specifying and Analyzing Transactional Consistency Models with Predicates

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A semantic and graph-theoretic foundation for transactional consistency with predicates

Item Operations vs. Predicate Operations

Read(x, v)

Write(x, v)

point access

returns one value

Employee

EID	Salary
X	8
Y	10
Z	6

PredRead(P, M)

$P : \text{Salary} < 9$

$M : \{X \mapsto 8, Z \mapsto 6\}$

condition-based access

returns a set M of **matched** values

What we know about transactional consistency models
without predicates

Review: Intuitive or Operational Specs

Eswaran et al., 1976

The Notions of Consistency and Predicate Locks in a Database System

K.P. Eswaran, J.N. Gray,
R.A. Lorie, and I.L. Traiger
IBM Research Laboratory
San Jose, California

SER

serial execution

Berenson et al., 1995

A Critique of ANSI SQL Isolation Levels

Hal Berenson
Phil Bernstein
Jim Gray
Jim Melton
Elizabeth O'Neil
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SI

snapshot + write-conflict control

Review: Axiomatic Specs

Cerone, Bernardi, and Gotsman, 2015

A Framework for Transactional Consistency Models with Atomic Visibility

Andrea Cerone, Giovanni Bernardi, and Alexey Gotsman

IMDEA Software Institute, Madrid, Spain

(VIS, AR)

Crooks et al., 2017

Seeing is Believing: A Client-Centric Specification of Database Isolation

Natacha Crooks

The University of Texas at Austin and Cornell University

Lorenzo Alvisi

The University of Texas at Austin and Cornell University

Yuer Pu

Cornell University

Allen Clement

Google, Inc.

Client-Centric Views

Biswas and Enea, 2019

On the Complexity of Checking Transactional Consistency

RANADEEP BISWAS, Université de Paris, IRIF, CNRS, France

CONSTANTIN ENEA, Université de Paris, IRIF, CNRS, France

Checking Complexity

Review: Dependency Graphs

Adya, 1999

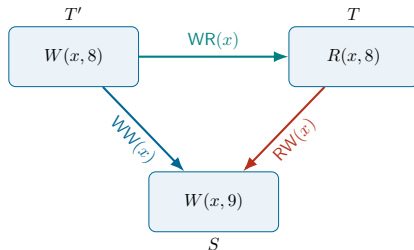
Weak Consistency: A Generalized Theory and Optimistic Implementations for Distributed Transactions

by

Atul Adya

Submitted to the Department of Electrical Engineering and Computer Science
in partial fulfillment of the requirements for the degree of

Doctor of Philosophy



Review: SER Characterization Theorem

Adya, 1999

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Bernstein, Hadzilacos, and Goodman, 1987

**Concurrency Control
and Recovery
in Database Systems**

P.A. BERNSTEIN • V. HADZILACOS • N. GOODMAN

Theorem (roughly stated)

*An item-operation history is serializable iff
its dependency graph is acyclic.*

Review: SI Characterization Theorem

Cerone and Gotsman, 2018

Analysing Snapshot Isolation

ANDREA CERONE, Imperial College London, UK
ALEXEY GOTSMAN, IMDEA Software Institute, Spain

Theorem (roughly stated)

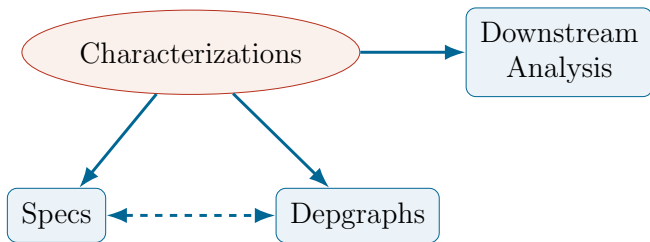
An item-operation history satisfies SI iff every cycle of its dependency graph has at least two adjacent anti-dependency (RW) edges.

Review: Downstream Analysis Tasks

Task	Representative work
History checking	Kingsbury and Alvaro, 2020; Tan et al., 2020; Huang et al., 2023; Cai et al., 2025
Protocol verification	Xiong, 2020
Program robustness	Bernardi and Gotsman, 2016

Review (Summary)

What we know about transactional consistency models
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What we know about transactional consistency models
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Review: Axiomatic Specs

Bouajjani, Enea, and Román-Calvo, 2025

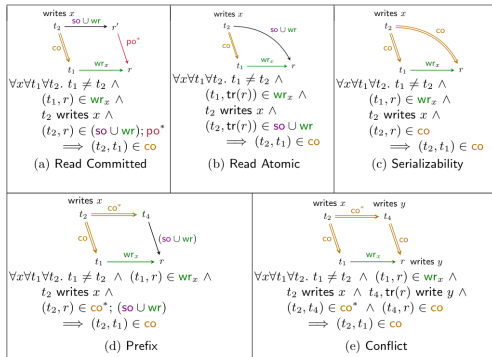
On the Complexity of Checking Mixed Isolation Levels for SQL Transactions

Ahmed Bouajjani¹, Constantin Enea², and Enrique Román-Calvo¹



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On checking complexity and algorithms; No dependency graph correspondence

Review: Dependency Graphs

Adya, 1999

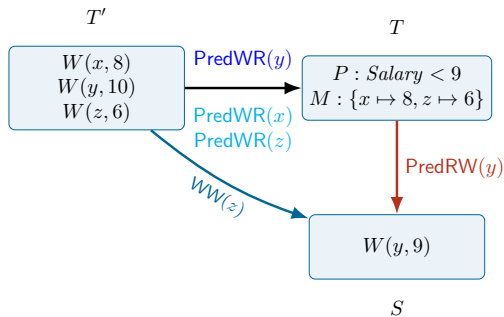
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T also “observes” **unmatched** $W(y, 10)$ of T'



S later than T' **changes matches** of T on y

Review: Dependency Graphs

Predicate-read-dependency, $T \xrightarrow{\text{PredWR}(x)} S$, definitions differ.

Predicate-anti-dependency, $T \xrightarrow{\text{PredRW}(x)} S$, definitions differ.
($\text{PredRW}^+ \subseteq \text{PredRW} \subseteq \text{PredRW}^-$)

PredRW

Adya, 1999

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S changes matches

PredRW⁺

Sun and Zou, 2025

Vbox: Efficient Black-Box Serializability Verification

Anonymous Author(s)

the “earliest” frontier S

PredRW⁻

Bouajjani, Enea, and Román-Calvo,
2025 (this work; CAV-inspired)

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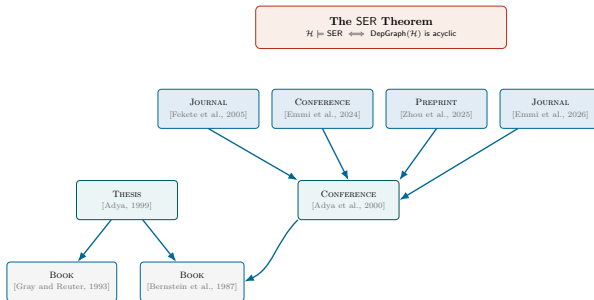


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S overwrites x ,
require no “change matches”

Review: SER Characterization



3.4 Correctness and Flexibility of the New Specifications

Our conditions for PL-3 provide the well-known notion of conflict-serializability [BHG87, GR93]. That is, $DSG(H)$ is acyclic, and G1a, and G1b are satisfied for a history H iff H is conflict-serializable. We can prove the equivalence of our PL-3 conditions with conflict-serializability using the proof given in [GR93]; our DSGs are similar to their graphs. This equivalence can also be proved along the lines of the proof given in [BHG87] for view-serializability.

AFAWK, no published rigorous proof in the literature

Review: SI Characterization

Adya, 1999

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Theorem 2: A history H consisting of committed transactions executes under Snapshot Isolation iff G1 and G-SI are disallowed.

G-SIa: Low-level *start-depends* $\prec_t$ relation

Definition 17: **Snapshot Read.** All reads performed by a transaction T_i occur at its start point. That is, if $r_i(x_j)$ occurs in history H , then:

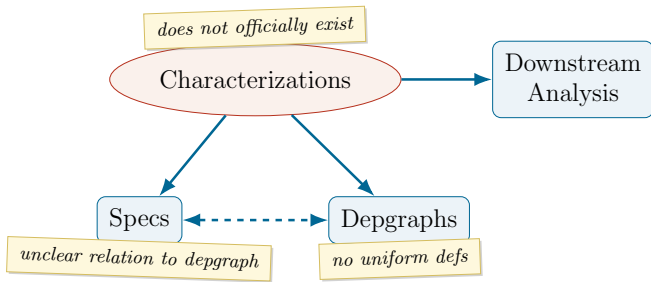
1. $c_j \prec_t s_i$, and
2. if $w_k(x_k)$ also occurs in H ($j \neq k$), then either
 - (a) $s_i \prec_t c_k$, or
 - (b) $c_k \prec_t s_i$ and $x_k \ll x_j$

Definition 18: **Snapshot Write.** If T_i and T_j are concurrent, they cannot both modify the same object. That is, if $w_i(x_i)$ and $w_j(x_j)$ both occur in history H , then either $c_i \prec_t s_j$ or $c_j \prec_t s_i$. This is the first-committer-wins property.

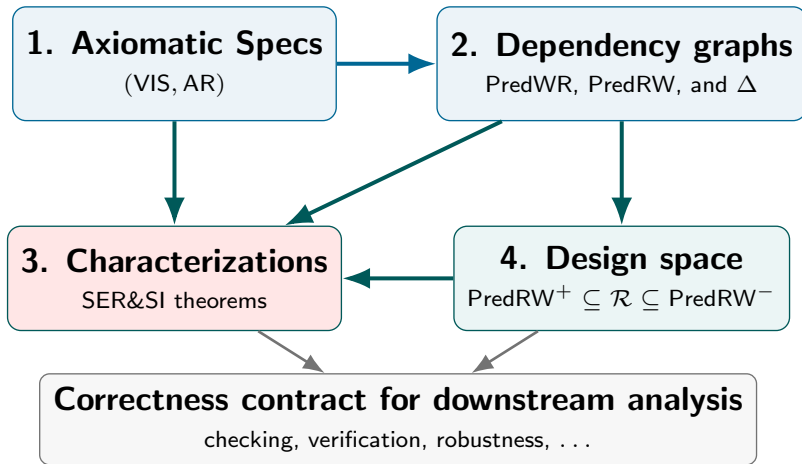
Proof based on *item-only* properties of SI

Review (Summary)

What we know about transactional consistency models
with predicates



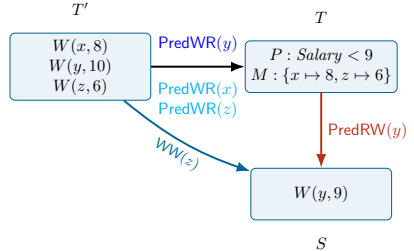
Contributions: Fill and Exploit the Gap



1. Axiomatic Specs

Table 2: Consistency axioms, constraining an abstract execution $(\mathcal{H}, \text{VIS}, \text{AR})$. (Generalized based on [11])

$\forall [E, \text{po}] \in \mathcal{T}, e \in E, x \in K, v \in V, F \subseteq E.$ $\left(\text{op}(e) = R(x, v) \wedge F = \{f \mid (\text{op}(f) = R(x, _) \vee \text{op}(f) = W(x, _)) \wedge f \xrightarrow{\text{po}} e\} \wedge F \neq \emptyset \right)$ $\implies \text{op}(\max F) = _ (x, v)$		(INT-ITEM)
$\forall [E, \text{po}] \in \mathcal{T}, e \in E, f \in E, x \in K, v \in V, P \in \mathcal{P}, M \subseteq K \times V, M' \subseteq K \times V, F \subseteq E, G \subseteq E.$ $\left(\text{op}(e) = \text{PR}(P, M) \wedge \text{op}(f) = \text{PR}(P, M') \wedge f \xrightarrow{\text{po}} e \wedge \neg(\exists g \in E. \text{op}(g) = \text{PR}(P, _) \wedge f \xrightarrow{\text{po}} g \xrightarrow{\text{po}} e) \wedge \right.$ $F = \{h \mid \text{op}(h) = W(x, _) \wedge f \xrightarrow{\text{po}} h \xrightarrow{\text{po}} e\}$ $\implies \left((F = \emptyset \implies ((x, v) \in M \iff (x, v) \in M')) \wedge \right.$ $\left. (F \neq \emptyset \implies ((x, v) \in M \iff (\text{op}(\max F) = W(x, v) \wedge P(x, v))) \right) \wedge$		(INT-PRED)
$\left(\text{op}(e) = \text{PR}(P, M) \wedge \neg(\exists g \in E. \text{op}(g) = \text{PR}(P, _) \wedge g \xrightarrow{\text{po}} e) \wedge \right.$ $G = \{h \mid \text{op}(h) = W(x, _) \wedge h \xrightarrow{\text{po}} e\} \wedge G \neq \emptyset$ $\implies ((x, v) \in M \iff (\text{op}(\max G) = W(x, v) \wedge P(x, v)))$		
$\forall T \in \mathcal{T}, x \in K, v \in V, T \vdash R(x, v) \implies \max_{\text{AR}} \{\text{VIS}^{-1}(T) \cap \text{WriteTx}_x\} \vdash W(x, v)$	(EXT-ITEM)	$\text{SO} \subseteq \text{VIS}$ (SESSION)
$\forall T \in \mathcal{T}, P \in \mathcal{P}, M \subseteq K \times V, x \in K, T \vdash \text{PR}(P, M, x) \implies$ $\boxed{x \in \text{DOM}(M) \implies \exists v \in V. (x, v) \in M \wedge \max_{\text{AR}} \{\text{VIS}^{-1}(T) \cap \text{WriteTx}_x\} \vdash W(x, v)} \wedge$ $\boxed{x \in K \setminus \text{DOM}(M) \implies \forall v \in V. (\max_{\text{AR}} \{\text{VIS}^{-1}(T) \cap \text{WriteTx}_x\} \vdash W(x, v) \implies \neg P(x, v))}$		
$\text{AR} ; \text{VIS} \subseteq \text{VIS}$ (PREFIX) $\text{VIS} = \text{AR}$ (TOTALVIS) $\forall S, T \in \mathcal{H}. S \models T \implies (S \xrightarrow{\text{VIS}} T \vee T \xrightarrow{\text{VIS}} S)$ (NOCONFLICT)		



External predicate consistency (EXT-PRED axiom)

For each externally observed key x , choose the **AR-latest** writer visible to the reader. Its value must equal the returned value when x **appears** in the result, and must fail the predicate when x is **omitted**.

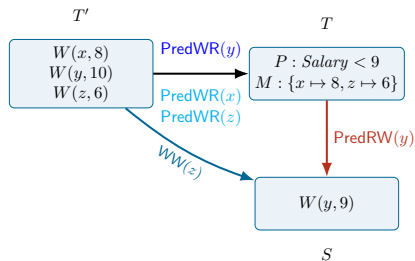
2. Dependency Graphs

Graph witness

$T' \xrightarrow{\text{PredWR}(y)} T$: T' supplies the version by which T externally observes y .

Observation-changing successor

$T \xrightarrow{\text{PredRW}(y)} S$: $T' \xrightarrow{\text{WW}(y)} S$ and S
changes the matches of T on y wrt T' .



3. Characterizations

Theorem (SER Characterization)

$\mathcal{H} \models \text{SER}$ iff it admits an *acyclic* predicate-aware depgraph.

Theorem (SI Characterization)

$\mathcal{H} \models \text{SI}$ iff it admits a predicate-aware depgraph where every cycle has two adjacent RW/PredRW edges.

3. Characterizations

Theorem (SER Characterization)

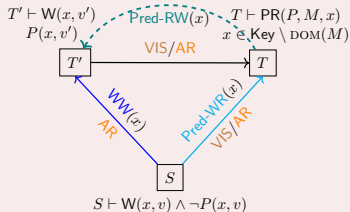
$\mathcal{H} \models \text{SER}$ iff it admits an *acyclic* predicate-aware depgraph.

Theorem (SI Characterization)

$\mathcal{H} \models \text{SI}$ iff it admits a predicate-aware depgraph where every cycle has two adjacent RW/PredRW edges.

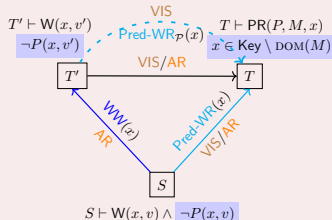
“if” direction: construct legal execution $\mathcal{A} \models \text{Ext-Pred}$ from graph

Challenge: mismatched witnesses



violation of $\text{EXT-PRED} \Rightarrow \text{PredRW} \Rightarrow$ forbidden cycle

Challenge (SI): mismatched witnesses



no need to recover the exact graph from execution $\mathcal{A}$

4. Design Space (of Predicate Anti-Dependencies)

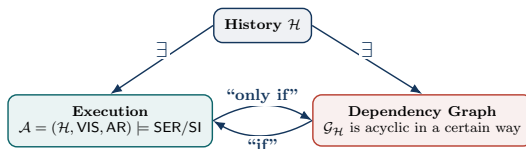
$$\text{PredRW}^+ \subseteq \mathcal{R} \subseteq \text{PredRW}^-$$

Every relation $\mathcal{R}$ preserves the same existential SER/SI characterization.

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$$\text{PredRW}^+ \subseteq \mathcal{R} \subseteq \text{PredRW}^-$$

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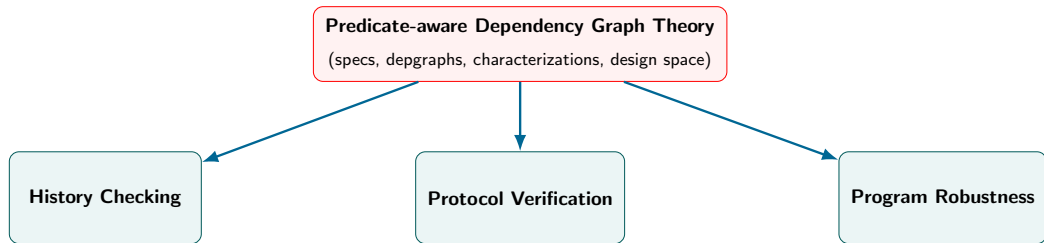
"if": $\text{PredRW}^+ \subseteq \mathcal{R}$

$\text{PredRW}^+ + \text{WW}$ recovers PredRW in reachability

"only if": $\mathcal{R} \subseteq \text{PredRW}^-$

$\text{PredWR} + \text{WW}$ suffices to derive safe AR orientation

Future Work: Downstream Analysis



[PredRW⁺, PredRW⁻]-guided history checker

- a lower persistent cycle can reject
- an upper acyclic graph can accept
- ambiguous cases are refined or solved

Takeaways: Theory Established, Downstream Analysis Enabled

1. **Specs.** Extend (VIS, AR) with item-wise external predicate observations.
2. **Dependency graphs.** Formally define PredRW with the “change matches” condition.
3. **SER/SI Characterizations.** Prove the same execution–graph equivalence style as the item-only results, while resolving the predicate-specific SI witness ambiguity.
4. **Proof-theoretic design space.** Every relation $\text{PredRW}^+ \subseteq \mathcal{R} \subseteq \text{PredRW}^-$ preserves the same existential history-level characterization.

Checking, Verification, Robustness, ...